

Week 1 Setup Lab

Lab Title: Getting Your Engineering Environment Running

Objective

Ensure your system is ready for the course and introduce basic signal visualization.

Part 1 — Environment Verification

- Run `sanity_check.py`
 - Take a screenshot of the plot
 - Submit screenshot
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Part 2 — Modify the Signal

Edit the script:

 Python



 Run

```
y = np.sin(2*x) + 0.5*np.random.randn(len(x))
```

Then:

- Plot the signal
 - Change frequency and noise level
-

Questions (Reflection)

- What effect does noise have on the signal?
 - How does increasing frequency change the plot?
 - What would make this data difficult to analyze?
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Submission

- Screenshot of plot

- Modified code
- Short reflection (3–5 sentences total)